INCH-POUND

MIL-C-24643/26C
14 March 1994
SUPERSEDING
MIL-C-24643/26B
3 October 1986

MILITARY SPECIFICATION SHEET

CABLE, ELECTRICAL, 600 VOLTS, TYPES LSDPS, LSFPS, LSTPS AND LS7PS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-24643.

REQUIREMENTS:

Qualification required.

Construction (watertight with circuit integrity)

	First	- Copper conductor, nickel coated (see table I for size and stranding).
	Second	- Extruded silicone rubber insulation (see table I for dimensions).
	Third	- Glass braid (see table I for dimensions). Standard identification code applied by method 4.
*	Fourth	- Transparent braid covering. The specified number of pairs (see table I) cabled together with a lay not greater than 10 inches. Fillers saturated with non-charring silicone based compound to form a firm, well-rounded assembly.
	Fifth	- A binder tape applied helically with overlap.
*	Sixth	- Silicone rubber jacket colored orange or red (see table I for thickness).
	Seventh	- Braided metal armor.

AMSC N/A FSC 6145

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TABLE I. Details.

Military part no. M24643/26	Type and Size	Conductors				Cable jacket thick-ness min avg (inch)	Overall diameter maximum (inch)	Cold bending manderl dia max (inches)	Conductor resistance (ohms)
		No. of conduc tors	AWG Size and Stranding	Diameter over insulation minimum (inches)	Diameter over braid covering maximum (inch)				
-01AN	LSDPS-3	2	16 (class B)	0.100	0.125	0.038	0.455	6.0	5.39
-02AN	LSDPS-4	2	14 (class B)	.116	.141	.038	.489	6.0	3.38
-03AN	LSDPS-6	2	12 (class B)	.143	.170	.050	.585	7.5	2.13
-04AN	LSDPS-9	2	10 (class B)	.166	.193	.050	.628	7.5	1.34
-05AN	LSDPS-14	2	9 (class B)	.200	.230	.063	.730	8.0	1.060
-06AN	LSTPS-3	3	16 (class B)	.100	.125	.038	.475	6.5	5.39
-07AN	LSTPS-4	3	14 (class B)	.116	.141	.050	.553	7.0	3.38
-08AN	LSTPS-6	3	12 (class B)	.143	.170	.050	.620	7.5	2.13
-09AN	LSTPS-9	3	10 (class B)	.166	.193	.050	.657	7.5	1.34
-10AN	LSTPS-14	3	9 (class B)	.200	.230	.063	.751	9.5	1.060
-11AN	LSTPS-23	3	7 (class C)	.257	.291	.063	.866	10.5	.643
-12AN	LSTPS-30	3	5 (class B)	.310	.348	.075	.989	12.0	.403
-13AN	LSFPS-14	4	9 (class B)	.200	.230	.063	.815	10.0	1.060
-14AN	LSTPS-6	7	12 (class B)	.143	.170	.063	.775	9.5	2.13
-15AN	LSTPS-14	7	9 (class B)	.200	.230	.063	.986	11.0	1.060

EXAMINATION AND TESTS:

	Requirements
Basic electrical:	
Conductor Resistance - ohms/1000 feet at 25°C, maximum	(see table I)
Voltage withstand - volts, root mean square, minimum	3000
Insulation resistance - megohms/1000 feet, minimum	
* Conductor to conductor	500
Conductor continuity	No failure
Group A:	
Visual and dimensional	No failure
The nickel coating on the conductors of the completed cable shall not be nicked, scraped or abraded so as to expose the copper conductor	No failure
Watertightness - see MIL-C-24643 for limits of water leakage	No failure

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EXAMINATION AND TESTS (continued):

	Requirements
Group B:	
* Cross-linked proof test (percent, maximum)	
Insulation	50
Jacket (When tested at 200°C)	50
Gas Flame - 1 hour at 230/130 volt root mean square	No failure
Drip - 95 ± 1°C	Zero
Cold bending, cable - (see table I for mandrel diameter)	No damage
Physicals (unaged)	
Insulation	
Tensile strength - lb/in^2, minimum	750
Elongation - percent, minimum	125
Jacket (cable)	
Tensile strength - lb/in^2, minimum	800
Elongation - percent, minimum	250
Durometer (type A) hardness, minimum	60
Group C:	
Physicals - cable aging (260 ± 10°C)	
Jacket	
Tensile strength - lb/in^2, minimum	600
Elongation - percent, minimum	150
*Cable sealant removability	No failure
*Group D:	
Flame propagation (cable)	No failure

QUALIFICATION INSPECTION:

* Qualification inspection shall include basic electrical, all of groups A, B, C and D, plus the following:

Gas Flame (3 hours) at 120/70 volts RMS	No failure
Aging and compatibility (cable) (125 ± 5°C)	No failure
For this test, omit all references to watertightness and fillers.	
Armor - conformance to material, construction and coverage	No failure
Specific gravity of extruded silicone insulation, maximum	1.55
Halogen content - percent, maximum	
Jacket	0.2
Fillers	0.2
Smoke index, maximum	
Jacket	35
* Fillers	45
Insulation	35
Toxicity index, maximum	
Jacket	5
Fillers	5
Insulation	1.5

UNIT ORDERING LENGTHS: All types and sizes 100 feet minimum.

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